



Variable Importance

VARIABLE IMPORTANCE

WHY DO WE NEED VARIABLE IMPORTANCE?

- Which variables drive predictions?
- Can we simplify the model?
- What insights can we extract?
- **Applications:**
 - Healthcare → risk factors
 - Finance → default drivers
 - Policy → key interventions

VARIABLE IMPORTANCE PRINCIPLES

Key Principle

- Variable importance is only meaningful if the model performs well
- Always check model performance first
- Poor model $\rightarrow$ misleading importance

General Setup

- Data: $\mathcal{D} = \{(x_i, y_i)\}$
 - Features: $x = (x_1, \dots, x_p)$
 - Model: $\hat{f}(x)$
 - Goal: quantify contribution of each feature
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LINEAR REGRESSION EXAMPLE

Setup

- $\hat{y} = \beta_0 + \sum_{j=1}^p \beta_j x_j$
- Coefficients represent feature effects
- Larger magnitude $\rightarrow$ more influence (with caveats)

Example

- $\hat{y} = 50,000 + 150x_1 + 20,000x_2 - 5,000x_3$
- sqft: 150
- bedrooms: 20,000
- age: -5,000

NOTE: Bedrooms appears most important (misleading)

FEATURE SCALING

- We scale **features**:

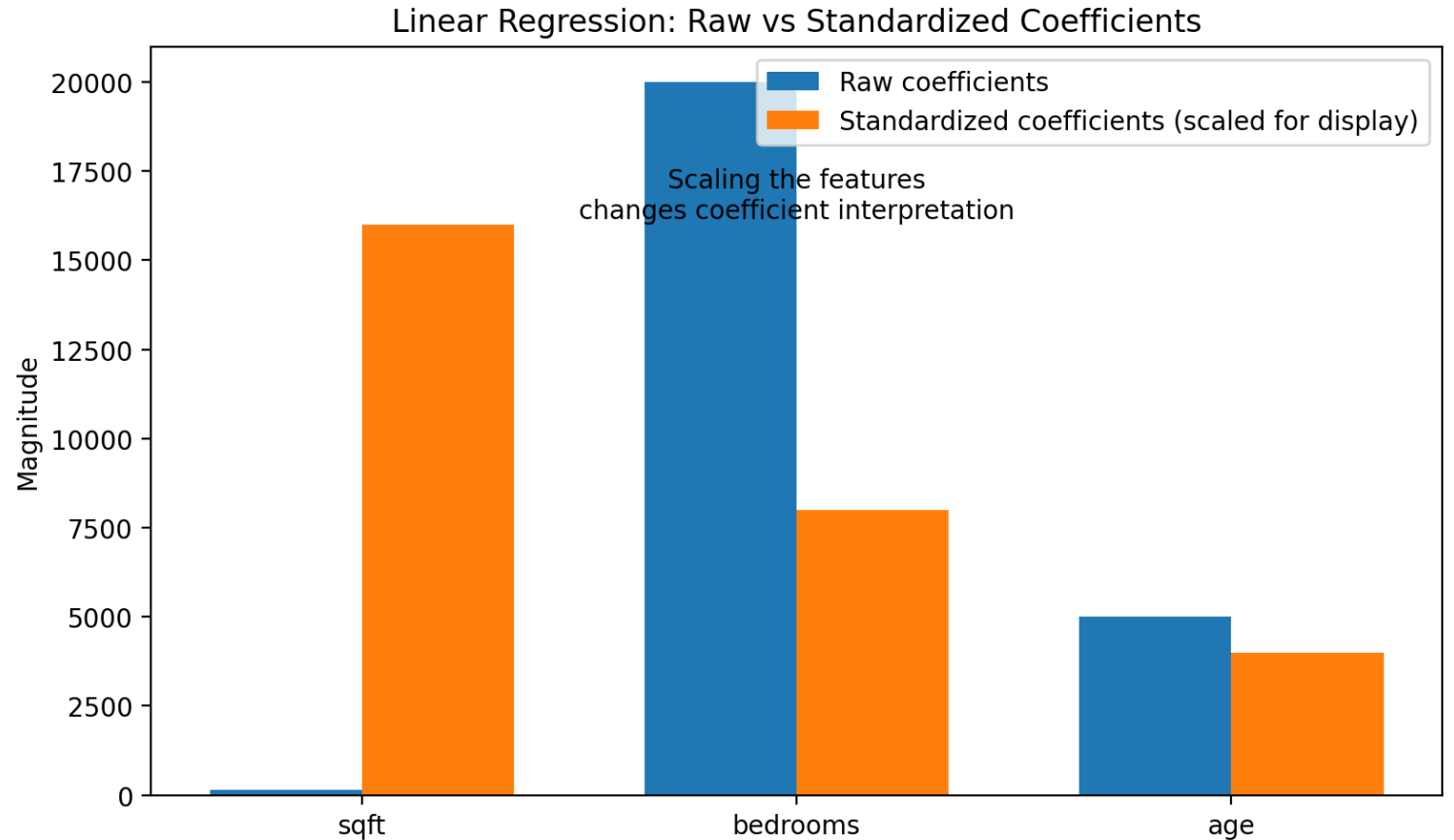
- $x_j^{(\text{std})} = \frac{x_j - \mu_j}{\sigma_j}$

- Then fit:

- $\hat{y} = \beta_0 + \sum \beta_j^{(\text{std})} x_j^{(\text{std})}$

- **Example (After Scaling)**

- sqft: **0.80**
 - bedrooms: 0.40
 - age: 0.20



This figure shows why raw coefficients can mislead us. Bedrooms looks largest in raw units, but once we account for feature scale, square footage becomes the strongest signal. The core lesson is that we should compare standardized effects, not raw coefficients, when ranking feature importance in linear models.

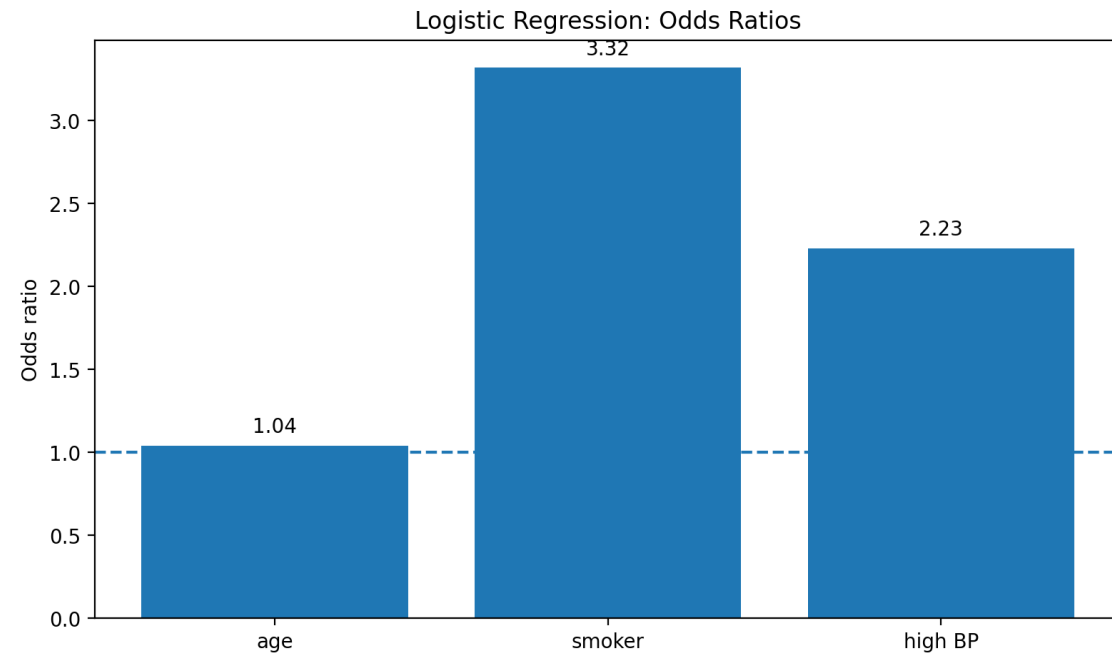
LOGISTIC REGRESSION

$$\log\left(\frac{p}{1-p}\right) = -5 + 0.04x_1 + 1.2x_2 + 0.8x_3$$

- x_1 :age
- x_2 :smoker
- x_3 :blood pressure

Interpretation

- Age: $e^{0.04} = 1.04$
- Smoking: $e^{1.2} = 3.32$
- Blood pressure: $e^{0.8} = 2.23$
- Smoking strongest predictor



"Here, odds ratios make logistic regression interpretable. A value near 1 means little change in odds, while larger values indicate stronger effects. Smoking has the largest odds ratio in this example, so it is the strongest predictor among these features."

LOGISTIC REGRESSION PITFALLS

Pitfalls:

- Log-odds interpretation
 - Scale issues
 - Class imbalance
 - Correlated predictors
 - Poor model performance
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DECISION TREE IMPORTANCE

Let:

- T : a decision tree
- t : a node in the tree
- x_j : feature j
- $I(t)$: impurity at node t
- n_t : number of samples at node t

- At a node t , suppose we split on feature x_j , producing:
 - left child: t_L
 - right child: t_R
- The **impurity decrease** from this split is:

$$\Delta I(t) = I(t) - \left(\frac{n_{t_L}}{n_t} I(t_L) + \frac{n_{t_R}}{n_t} I(t_R) \right)$$

IMPURITY FUNCTIONS

- **Classification (Gini impurity)**

- $I(t) = 1 - \sum_{k=1}^K p_k(t)^2$

- where $p_k(t)$ is the proportion of class k in node t .

- **Regression (Variance / MSE)**

- $I(t) = \frac{1}{n_t} \sum_{i \in t} (y_i - \bar{y}_t)^2$

The importance of feature x_j is the **total impurity reduction contributed by splits on that feature**:

$$VI_j^{tree} = \sum_{t \in T: \text{split on } x_j} \frac{n_t}{n} \Delta I(t)$$

where:

- $\frac{n_t}{n}$: weights splits by how many samples reach node t
 - n : total number of samples
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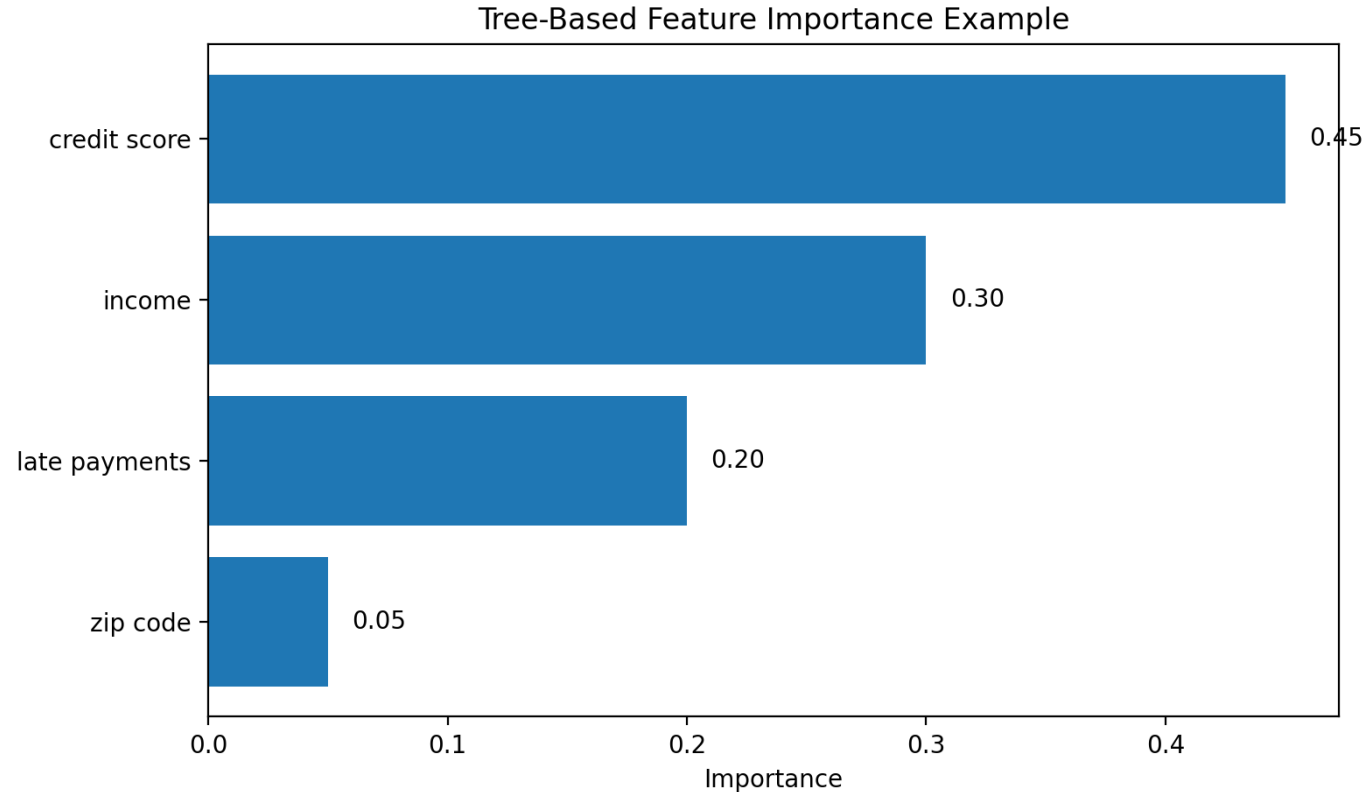
RANDOM FORESTS

Mean Decrease in Impurity (MDI)

- The standard random forest importance (used in CART-based forests) is:

$$VI_j^{RF} = \frac{1}{B} \sum_{b=1}^B VI_j^{(b)}$$

RF IMPORTANCE EXAMPLE



This tree-based importance chart summarizes total impurity reduction by feature. Credit score contributes the most, followed by income and late payments. Remember that this ranking is model-specific and can be affected by feature correlation or split bias.

DT AND RF MDI LIMITATIONS

1. Bias Toward High-Cardinality Features

- Features with many split points:
- More chances to reduce impurity
- Artificially inflated importance

2. Correlated Features

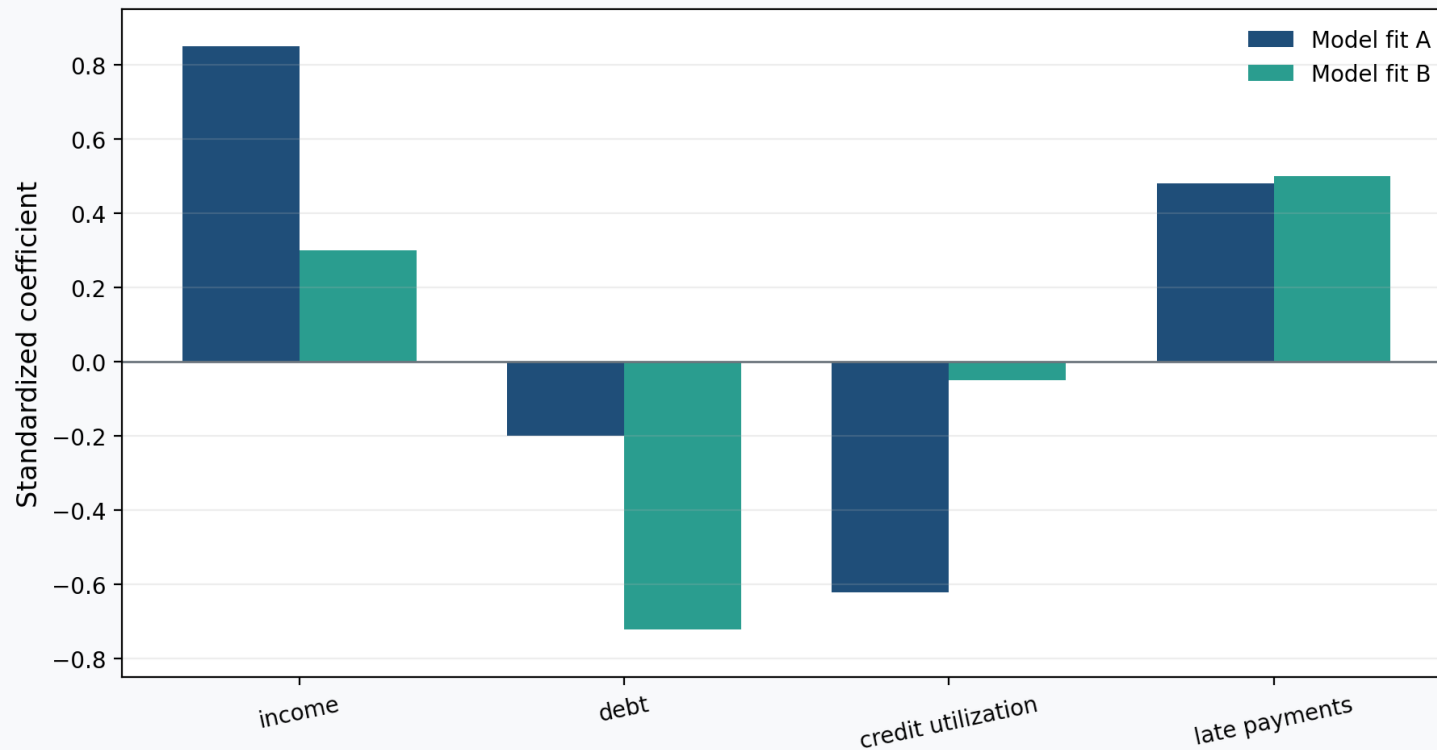
- If $x_1 \approx x_2$:
- Trees may split on either
- Importance is **shared or diluted**

3. Training-Based Measure

- **training data impurity reduction**
 - NOTE: Not directly tied to generalization performance

CORRELATED FEATURES

Multicollinearity: Importance Can Move Between Correlated Features



With correlated predictors, coefficients can shift between model fits even when overall performance is similar. The predictive signal is shared, so individual coefficient-based importance becomes unstable. Interpret these values cautiously under multicollinearity.

MODEL AGNOSTIC

- The above-described mechanisms are model specific.
- What about other models, such as ANNs?

PERMUTATION IMPORTANCE

Measure how much model performance degrades when a feature's information is destroyed.

- $\hat{f}$:trained model
- $L(\cdot)$:loss (or negative performance metric)
- X : original data
- $X_{perm(j)}$:data with feature x_j randomly shuffled

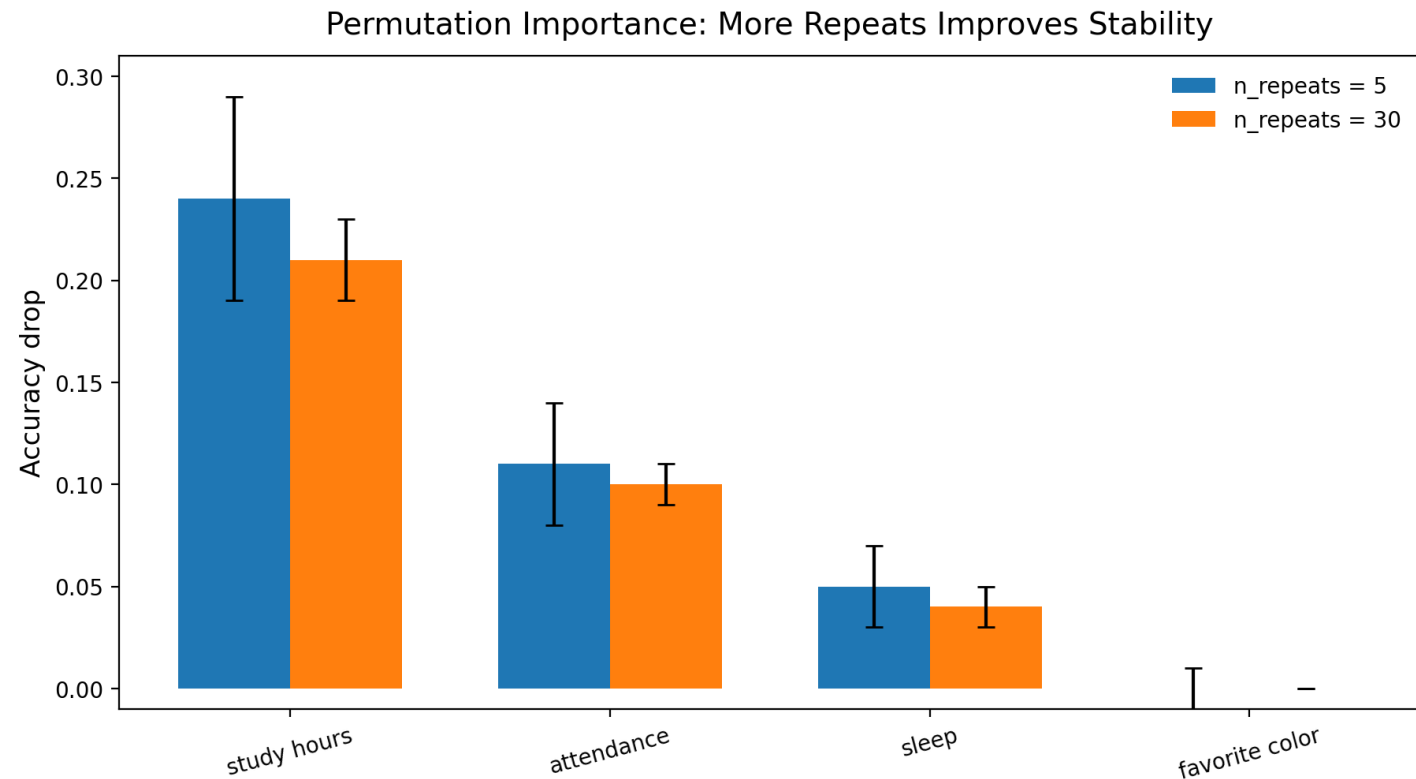
$$VI_j = L(\hat{f}(X_{perm(j)}), Y) - L(\hat{f}(X), Y)$$

PERMUTATION IMPORTANCE

Algorithm (scikit-learn style)

- For each feature x_j :
- Compute baseline score:
- S_{orig} (Score)
- Repeat for $r = 1, \dots, R$:
 - Shuffle values of x_j
 - Compute score $S_j^{(r)}$
- Estimate importance:

$$VI_j = \frac{1}{R} \sum_{r=1}^R (S_{orig} - S_j^{(r)})$$



Permutation importance asks: How much does model accuracy drop when one feature is shuffled? Study hours causes the largest drop, so it is most important. The error bars show variation across repeats, which helps us separate stable signals from noise.